VIVEKANAND EDUCATION SOCIETY
INSTITUTE OF TECHNOLOGY
(AUTONOMOUS)

Department of Computer Engineering

CAPSTONE PROJECT LOGBOOK

(26CMCP71 : Capstone Project - II)

2026-27

GROUP MEMBERS

1. Sakshi Bhansali (7)
2. Shreya Gokhale (19)
3. Srushti Mahadik (40)
4. Rochelle Ann Teddy (50)

Project Mentor

Mrs.Pallavi Saindane - Assistant Professor

VIVEKANAND EDUCATION SOCIETY
INSTITUTE OF TECHNOLOGY (AUTONOMOUS)

Department of Computer Engineering

INSTITUTE VISION & MISSION

VISION:

To build a vibrant, knowledge-driven ecosystem that promotes holistic learning, inspires innovation, practices sustainability and value-based education, and shapes socially conscious, industry-ready technocrats.

MISSION:

- To inculcate a culture of multidisciplinary value based education.
- To inspire students to apply their knowledge to meet the needs of industry and society.
- To foster research, entrepreneurship and lifelong learning for societal growth

COMPUTER ENGINEERING DEPARTMENT

VISION:

To establish a center of excellence in computing by empowering students with advanced technological skills for fostering innovation, research, and societal impact.

MISSION:

- To cultivate a teaching learning environment that unites theory with computing skills, empowering students to adapt to evolving technologies.
- To strengthen core computing knowledge by fostering research, innovation, and entrepreneurship through multi-disciplinary learning.
- To prepare competent professionals with industry-ready skills, encouraging social awareness, sustainable development, and a commitment to lifelong learning.

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PROGRAM EDUCATIONAL OBJECTIVES (PEO's)

I	To equip Computer Engineering students with a strong foundation in mathematics, science, and core computing and engineering principles, enabling them to comprehend, analyze, and design effective technology-driven solutions.
II	To foster a multidisciplinary mindset in Computer Engineering students through flexible learning, enabling adaptability along with strong communication, teamwork, and leadership skills.
III	To empower Computer Engineering students with advanced computing and software skills, fostering innovation, entrepreneurship, and sustainable practices for lifelong learning.
IV	To provide Computer Engineering students with broad educational and research experiences through academic and industrial collaboration programs, preparing them for industry-ready careers.

PROGRAM OUTCOMES (POs)

Program Outcome Code	Program Outcome Description
PO1	**Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop the solution of complex engineering problems.
PO2	**Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
PO3	**Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
PO4	**Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.(WK8).

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PO5	**Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
PO6	**The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.(WK1, WK5, and WK7).
PO7	**Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
PO8	**Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
PO9	**Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural,language, and learning differences.
PO10	**Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
PO11	**Life-Long Learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1	**Professional Skills -** The ability to develop programs for computer based systems of varying complexity and domains using standard practices.
PSO2	**Successful Career -** The ability to adopt skills, languages, environment and platforms for creating innovative career paths, being successful entrepreneurs or for pursuing higher studies.

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STUDENT INFORMATION

Project Title: AstroSense - Exoplanet Vetting on Kepler Lightcurves

SDGs Mapped: SDG 4 - Quality Education, SDG 9 - Industry, Innovation and Infrastructure

	Student 1	Student 2	Student 3	Student 4
Roll No.	7	19	40	50
Name	Sakshi Bhansali	Shreya Gokhale	Srushti Mahadik	Rochelle Ann Teddy
Class with Division	D17A	D17A	D17A	D17A
Contact No.	9969397971	7977081818	9619911930	8657845294
E-mail	2023.sakshi.bhansali@ves.ac.in	2023.shreya.gokhale@ves.ac.in	2023.srushti.mahadik@ves.ac.in	2023.rochelle.teddy@ves.ac.in
Address	104-A, Vardhaman Park CHS	B-4 Mahavir APT	Acacia-604,Ashar Maple	1004 A1 Trinity Paradise
	Sector 17, Vashi	Ramnagar, Shiv Mandir Rd	Ghatipada Road, Vaishalinagar	Kalyan - Shilphata Rd, Shilphata
	Navi Mumbai	Dombivli	Mulund,Mumbai	Thane, Maharashtra
	400703	421201	400080	421204

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INSTRUCTIONS TO STUDENTS:

1. The logbook must be submitted to the Project Mentor or Project Coordinator for verification and evaluation of project activities at least once in a week.
2. A logbook duly signed by the project mentor must be submitted with a project blackbook for evaluation at the end of semester to the department.

DECLARATION

I declare that this project represents my ideas in my own words and wherever others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my project work. I promise to maintain minimum 75% attendance, as per the Institute norms. I understand that any violation of the above will be cause for disciplinary action by the Institute.

Yours Faithfully

1. Sakshi Bhansali (7)
2. Shreya Gokhale (19)
3. Srushti Mahadik (40)
4. Rochelle Ann Teddy (50)

(Signature of Students)

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Letter of Acceptance

I undersigned, **Prof.** ***Mrs. Pallavi Saindane*** working in the Computer Engineering department, willing to guide the Project titled ***Astrosense - Exoplannet Vetting on Kepler Lightcurves*** for the Capstone Project - II Semester VII respectively for the ***Academic Year 2026-27.*** The names of the students are:

1. **Sakshi Bhansali (7)**
2. **Shreya Gokhale (19)**
3. **Srushti Mahadik (40)**
4. **Rochelle Ann Teddy (50)**

[redacted]	[redacted]	[redacted]
(Project Mentor)	(Capstone Project Coordinator)	(HOD Department of Computer Engineering)

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COURSE OUTCOMES FOR CAPSTONE PROJECT–II

Course Outcome	Description	PO's Covered	PSO's Covered
CO1	Implement and integrate all modules of the proposed system effectively.	PO1, PO2, PO3, PO5	PSO1, PSO2
CO2	Perform testing, debugging, validation, and performance optimization of the developed solution.	PO2, PO3, PO4, PO5	PSO1, PSO2
CO3	Analyze project outcomes using appropriate engineering tools and metrics.	PO2, PO4, PO5	PSO1, PSO2
CO4	Demonstrate deployment readiness, usability, security, scalability, and sustainability considerations.	PO3, PO5, PO6, PO7	PSO1, PSO2
CO5	Analyze, interpret, and present research outcomes effectively using appropriate research methodologies, analytical tools, and presentation techniques.	PO2, PO4, PO5, PO8, PO10	PSO1, PSO2
CO6	Present and defend the complete project professionally through demonstrations, presentations, and comprehensive technical documentation.	PO8, PO9, PO10, PO11	PSO1, PSO2

CO–PO–PSO Mapping Table

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	3	3	–	3	–	–	–	3	–	2	3	3
CO2	2	3	2	3	3	–	–	–	2	–	2	3	2
CO3	2	3	2	3	3	–	–	–	–	2	2	3	2
CO4	2	2	3	2	3	2	3	2	–	–	3	3	3
CO5	2	3	–	3	3	–	–	2	–	3	2	2	2
CO6	–	–	–	–	2	–	–	2	3	3	2	2	2

Mapping scale: 3 = High correlation; 2 = Moderate correlation; 1 = Low correlation; – = No direct correlation.

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SCHEDULE FOR CAPSTONE PROJECT - II

Date	Week	Contents	Remark	Guide Sign
10/08/26	1	Developed a reliable Kepler data acquisition pipeline with initial data validation and quality checks.	Proceed with preprocessing.	
17/08/26	2	Implemented light-curve preprocessing, cleaning, and standardization for model development.	Good, proceed further.	
24/08/26	3	Explored BATMAN-based synthetic transit injection for simulating exoplanet transit signals.	Injection approach tested.	
31/08/26	4	Developed and evaluated the Windowed 1D CNN model for exoplanet detection.	CNN results reviewed	
07/09/26	5	Implemented the Attention-MIL CNN model for star-level classification using attention-weighted window aggregation.	Evaluate and refine results	
21/09/26	6	Studied State Space Models (SSMs) and their applications in long-sequence astronomical time-series modeling	Explore SSM approaches.	
28/09/26	7	Explored selective SSM architectures, particularly Mamba-2, and their suitability for Kepler light-curve analysis.	Begin ssm implementation.	
05/10/26	8	Implemented preliminary Mamba-2 variants and investigated different input configurations and synthetic data experiments.	Refine model variants.	
07/10/26	9	Refined the Mamba-2 methodology by transitioning to real-data-only experiments. Developed V5 model variants using flux-only and flux-plus-time-gap representations with controlled training and evaluation.	Continue real-data evaluation.	
09/10/26	10	Conducted multi-seed Mamba-2 experiments, ensemble evaluation, and comparison with machine learning baselines.	Validate comparative results.	
12/09/26	11	Analyzed SSM performance, investigated model limitations, and consolidated CNN and Mamba-2 experimental findings.	Proceed with system integration.	

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19/09/26	12	Integrated models into the UI, implemented result visualizations and exportable reports, and completed system testing and documentation.	Finalize report and documentation.	

PROGRESS / ATTENDANCE REPORT

Title of the Project:	
Group No.	**Sakshi Bhansali** **Shreya Gokhale** **Srushti Mahadik** **Rochelle Ann Teddy**
Name of the Project Mentor: Mrs Pallavi Saindane	

Sr. No	Date	Attendance				Progress/Suggestion	Mapping		
		1	2	3	4		CO	PO	PSO
1	10/08/26	✓	✓	✓	✓	Project scope and data acquisition reviewed.	CO5	PO1, PO2, PO4, PO5, PO8, PO10, PO11	PSO1, PSO2
2	17/08/26	✓	✓	✓	✓	Preprocessing progress reviewed.	CO5	PO1, PO2, PO4, PO5, PO8, PO10, PO11	PSO1, PSO2
3	24/08/26	✓	✓	✓	✓	Synthetic data generation approach discussed.	CO5	PO1, PO2, PO4, PO5, PO8, PO10, PO11	PSO1, PSO2
4	31/08/26	✓	✓	✓	✓	CNN implementation reviewed.	CO1, CO2	PO1, PO2, PO3, PO4, PO5, PO9, PO11	PSO1, PSO2
5	07/09/26	✓	✓	✓	✓	Attention-based model reviewed.	CO1, CO2	PO1, PO2, PO3, PO4, PO5, PO9, PO11	PSO1, PSO2
6	21/09/26	✓	✓	✓	✓	SSM concepts and research progress	CO5	PO1, PO2,	PSO1,

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						discussed.		PO4, PO5, PO8, PO10, PO11	PSO2
7	28/09/26	✓	✓	✓	✓	Mamba-2 architecture reviewed.	CO5	PO1, PO2, PO4, PO5, PO8, PO10, PO11	PSO1, PSO2
8	05/10/26	✓	✓	✓	✓	Preliminary implementation evaluated.	CO1, CO2	PO1, PO2, PO3, PO4, PO5, PO9, PO11	PSO1, PSO2
9	07/10/26	✓	✓	✓	✓	Model refinement discussed.	CO1, CO2, CO3	PO1, PO2, PO3, PO4, PO5, PO9, PO10, PO11	PSO1, PSO2
10	09/10/26	✓	✓	✓	✓	Experimental results reviewed.	CO2, CO3, CO5	PO1, PO2, PO3, PO4, PO5, PO8, PO9, PO10, PO11	PSO1, PSO2
11	12/10/26	✓	✓	✓	✓	Performance analysis discussed.	CO2, CO3, CO5	PO1, PO2, PO3, PO4, PO5, PO8, PO9, PO10, PO11	PSO1, PSO2
12	19/10/26	✓	✓	✓	✓	Final implementation and documentation reviewed.	CO1, CO4, CO6	PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11	PSO1, PSO2

Sign of the Project Mentor

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EXAMINER'S FEEDBACK FORM

Name of External examiner:_____

College of External examiner:_____

Name of Internal examiner:_____

Date of Examination:_____/____/_____

No. of students in project team:

Availability of separate lab for the project: Yes / No

Student Performance Analysis (Put Tick as per your Observation)

	Excellent (3) Very Good (2) Good (1)			
Sr. No.	**Observation**	**(3)**	**(2)**	**(1)**
1	Quality of problem and Complex Engineering Problems			
2	Innovativeness in solutions			
3	Cost effectiveness and Societal impact			
4	Full functioning of working model as per stated requirements			
5	Effective use of skill sets			
6	Effective use of standard engineering norms			
7	Contribution of an individual as member or leader			
8	Clarity in written and oral communication			
9	Overall performance			

Suggest new Innovative Technique/Idea/ objectives related to this project.

[redacted]

Signature of External Examiner **Signature of Internal Examiner**